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Does the partition "Type of s1" satisfy the completeness property? If not, give a value for s1 that does not fit in any block.

I believe it does fit the completeness property. The value of null is covered, the empty set is covered, and the only other possibility is a set with elements and that is covered.

Does the partition "Type of s1" satisfy the disjointness property? If not, give a value for s1 that fits in more than one block.

Yes I believe it does satisfy the disjointness property. Null does not overlap with empty set {} or a set containing at least one element, in simple terms none of them overlap with each other.

If necessary, fix "Type of s1".

Not necessary

Does the partition "Relation between s1 and s2" satisfy the completeness property? If not, give a pair of values for s1 and s2 that does not fit in any block.

Yes I believe it does satisfy the completeness property, If the sets are the same its covered, if the sets have some elements in common its covered, if it has no elements in common its covered, and the empty set is covered.

Does the partition "Relation between s1 and s2" satisfy the disjointness property? If not, give a pair of values for s1 and s2 that fits in more than one block.

No I don't think it satisfies the disjointness property. For example if s1 and s2 both = empty set {}, then it satisfies all blocks, they are both the same set, empty set is a subset of an empty set, and they have no elements in common because logically they have no elements at all.

If necessary, fix "Relation between s1 and s2".

s1 and s2 represent the same set

s1 is a proper subset of s2

S2 is a proper subset of s1

s1 and s2 are non-empty and have no elements in common